EMPIRICAL EVIDENCE FOR THE SHARP CONJECTURE B4 AT $d = 4$ AND THE UNIVERSAL ξ_0-IDENTITY $\xi_0(b) = d/\beta_d^{1/d}$

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Abstract. We extend the PCF-2 v1.1 quartic test of Conjecture B4 (PCF-2 v1.1, Sessions A2/B/C1) from $d = 3$ to $d = 4$ on a lex-window catalogue of 60 Z-primitive irreducible quartics. The empirical A-distribution is unimodal at $A = 8$: 60/60 families lie within 0.10 of $A = 2d = 8$, mean $\overline{A} = 7.954$, $\sigma_A = 0.0037$. We additionally derive and verify a universal Newton-polygon characteristic root $\xi_0(b) = d/\beta_d^{1/d}$ at $d = 4$, recovering the proven $d = 2$ identity $\xi_0 = 2/\sqrt{\beta_2}$ and supplying *op:xi0-degree-d* from channel-theory-v11. Together, the two results refine *op:d2-anomaly*: the $\{2d-1, 2d\}$ splitting at $d = 2$ is so far the unique multi-branch instance, with $d \in \{3, 4\}$ both populating only the sharp branch $A = 2d$ within the lex-window catalogue.

1. Setup and conjectures

Recall PCF-2 v1.1, Conjecture B4 (sharp form, Session B):

Conjecture 1 (B4, sharp)**.** *For every Z-primitive irreducible $b(n) \in \mathbb{Z}[n]$ of degree $d \geq 3$ with $a_n = 1$, the WKB-decay exponent of the partial-tail errors $\delta_n = L_n - L$ satisfies $A = 2d$ in the asymptotic ansatz $\log|\delta_n| \sim -A\, n \log n + \alpha n - \beta \log n + \gamma$.*

At $d = 2$ Conjecture B4 fails as stated: PCF-1 v1.3 Theorem 5 proves the split $A \in \{2d-1, 2d\} = \{3, 4\}$ driven by $\mathrm{sgn}(\Delta_2)$.

The companion conjecture introduced in channel-theory-v11, *op:xi0-degree-d*, asks whether the proven $d = 2$ identity $\xi_0(b) = 2/\sqrt{\beta_2}$ (Proposition 3.3.A) extends to all $d \geq 1$ in the form $\xi_0(b) = c(d)/\beta_d^{1/d}$.

2. Catalogue

We enumerate $b(n) = a_4 n^4 + a_3 n^3 + a_2 n^2 + a_1 n + a_0$ with $a_4 \in \{1, 2, 3, 5, 7\}$, $a_{0,1,2,3} \in \{-3, \ldots, 3\}$ in lex order $(a_4, a_3, a_2, a_1, a_0)$, keeping gcd $= 1$ and irreducible over $\mathbb{Q}$, until 60 entries. Galois group distribution: $57 \times S_4 + 3 \times D_4 + 0 \times V_4/C_4/A_4$. Real signature: $(r_1, r_2) = (2, 1)$ for 59, $(4, 0)$ for 1. The catalogue contains no CM-quartic ($r_1 = 0$). The window is dominated by mixed S_4; this is a structural fact about the lex window and not a sampling artefact.

Calibration anchors $x^4 - 2$ (D_4), $x^4 + 1$ (V_4), $x^4 - x - 1$ (S_4) all reproduce expected Galois group.

3. Empirical evidence for Conjecture B4 at $d = 4$

For each cataloged family we compute L_b at dps $= 1200$ for $N \in [10, 60]$ step 2, with reference $N_{\mathrm{ref}} = 150$, and fit the ansatz of Conjecture 1 by linear LSQ in $(\log N \cdot N,\ N,\ \log N,\ 1)$. The protocol matches Session B's $d = 3$ harvest with adjusted window $[10, 60]$ (vs. $[10, 100]$) because at $d = 4$ the $A\, N \log N$ decay is faster and $N = 60$ already yields $|\delta_N| \sim 10^{-700}$ within the working precision floor.

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Results (Phase Q1-2).

- Verdict counts: `b4_consistent_at_8` : 60/60; `b4_split_candidate_at_7` : 0; `b4_violating` : 0.
- Empirical A-distribution: $\overline{A} = 7.9544$, $\sigma_A = 0.0037$, $A_{\min} = 7.9488$, $A_{\max} = 7.9605$.
- All 60 families satisfy $|A_{\text{fit}} - 8| \leq 0.0512 < 0.10$ and $|A_{\text{fit}} - 7| \geq 0.95 \gg 0.30$.
- Stderr range [0.0021, 0.0028] uniformly.

Remark 2 (Systematic offset)**.** *The cluster $\overline{A} = 7.954$ sits 0.046 below $A = 8$. The same 4-parameter ansatz at $d = 3$ produced $\overline{A} = 5.973$ ($\sigma = 0.026$, Session B), at $d = 2$ produced $\overline{A} = 3.97$ on the $A = 4$ stratum (PCF-1 v1.3). The offset scales roughly as 0.05 per unit increase in d and is consistent with a fit-window bias from the absence of a $-(\log\log N)$ correction in the ansatz. None of the 60 families in this catalogue is thereby pushed toward $A = 7$.*

4. Universal ξ_0-identity at $d = 4$

For a degree-d PCF with $a_n = 1$ and $b(n) = \sum_{k=0}^{d} \beta_k n^k$, the partial-denominator generating function $f(z) = \sum_n Q_n z^n$ satisfies

$$L_d\, f \;=\; (\text{polynomial in } z \text{ of degree } \leq 1), \qquad L_d \;=\; 1 - z\, B_d(\theta + 1) - z^2, \quad \theta = z\partial_z,$$

where $B_d(t) = \sum_{k=0}^{d} \beta_k\, t^k$.

Proposition 3 (Universal ξ_0-identity, derived)**.** *The Newton polygon of L_d at $z = 0$ has a unique nontrivial slope-$(1/d)$ edge $E : (0, 0) \to (1, d)$. The characteristic polynomial along E under the WKB ansatz $f \sim \exp(c/u)$, $z = u^d$, is*

$$\chi_d(c) \;=\; 1 \;-\; \frac{\beta_d}{d^d}\, c^d.$$

Its leading positive real root is

$$\boxed{\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}} \;=\; c(d)/\beta_d^{1/d} \text{ with } c(d) = d}.$$

Proof. $\theta \exp(c/u) = (u/d)\, \partial_u \exp(c/u) = -(c/(du)) \exp(c/u)$. So θ^j acts as $(-c/(du))^j$ at leading order. The single edge E of L_d at slope $1/d$ joins $(0, 0)$ (coefficient $+1$) and $(1, d)$ (coefficient $-\beta_d$). Substituting $z = u^d$ and collecting the u^0 coefficient of $L_d \exp(c/u)$ gives $1 - \beta_d(-c)^d/d^d = 0$, equivalently $1 = \beta_d c^d/d^d$. The positive real root is $c = d/\beta_d^{1/d}$. □

Numerical verification at $d = 4$ (Phase Q1-3). On 8 representative quartics — one per available bin (`mixed_S4` smallest $|\Delta_4|$, `mixed_S4` largest $|\Delta_4|$, `mixed_D4`, `+_real_D4`), three out-of-window anchors ($x^4 - 2,\ x^4 + 1,\ x^4 - x - 1$), and one $a_4 = 7$ sample — the inferred constant $c(4) := \xi_0(b)\, a_4^{1/4}$ takes the value 4 to spread 0 at dps = 80. The conjecture is verified at $d = 4$ with $c(4) = d = 4$, as predicted by Proposition 3.

5. Cross-degree compilation and refined op:d2-anomaly

d	A branches predicted	A branches empirical	$\xi_0(b)$	source
2	$\{3, 4\}$	$\{3, 4\}$ split by $\mathrm{sgn}(\Delta_2)$	$2/\sqrt{\beta_2}$	PCF-1 v1.3 Thm 5
3	$\{5, 6\}$	$\{6\}$ alone (50/50)	$3/\beta_3^{1/3}$ (predicted)	PCF-2 v1.1
4	$\{7, 8\}$	$\{8\}$ alone (60/60)	$4/\beta_4^{1/4}$ (verified)	this note

Refined op:d2-anomaly status. The $\{2d-1, 2d\}$ split at $d = 2$ is so far the unique populated multi-branch instance. Both $d = 3$ (Session B/C1) and $d = 4$ (this note) populate only the sharp branch $A = 2d$ within their respective lex windows.
Caveats. (i) The $d = 4$ catalogue contains no CM quartic ($r_1 = 0$, V_4/C_4); a CM-quartic harvest is the natural follow-up. (ii) The $\overline{A} \approx 2d - 0.05$ cluster is fit-window bias, not a genuine displacement (Remark 2). (iii) Universality of ξ_0 is an *operational* (Newton-polygon) statement, not yet a Borel-radius identity; the latter remains open per CC-PIPELINE-G / BOREL-CHANNEL-5X.

6. Conjecture B4 at $d \geq 3$: status after Q1

Combining Sessions B/C1 ($d = 3$) and Q1 ($d = 4$): 50/50 cubic + 60/60 quartic = 110/110 families verdict $A = 2d$, *no* $A = 2d - 1$ *branch detected* within the joint lex window. Conjecture 1 (B4 sharp form for $d \geq 3$) remains the working program statement; B4′ (the conjectured $A = 2d-1$ elementary-positive branch) is empirically empty in both degrees.

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